

CO2 AT2

NAME: A K Selva Prabhu

REG NO: 192421224

COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

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The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The assignment is titled "CO2-AT2 Assignment". On the left, a "QUESTIONS" sidebar lists three questions, with the first one, "Q1 Bank Token Management System", selected. The main area displays the question details: "Customers receive tokens in the order they arrive at a bank counter." and "Requirements: Add customer tokens to a Queue, Serve customers in FIFO order, Display the next customer to be served and the remaining queue". Below this, the "Your Code" section shows a Java program using a LinkedList to manage a queue of customer IDs. The "Input" section has a text box for "stdin input". The "Output" section shows the program's output: "Queue : [A101, A102, A103], Served : A101, Next Customer : A102, Remaining Queue : [A102, A103]".

SIMATS Online Programming Lab
JAVA PROGRAMMING

CO2-AT2 Assignment

QUESTIONS

- Q1 Bank Token Management System 10 marks
- Q2 Railway Coach and Seat Parser 10 marks
- Q3 Employee ID Normalization System 10 marks

3/3 answered

Q1 Bank Token Management System 10 marks

Customers receive tokens in the order they arrive at a bank counter.

Requirements:

- Add customer tokens to a Queue
- Serve customers in FIFO order
- Display the next customer to be served and the remaining queue

Your Code

```
1 import java.util.*;
2
3 class Bank {
4
5     public static void main(String args[]) {
6
7         Queue<String> q = new LinkedList<>();
8
9         q.add("A101");
10        q.add("A102");
11        q.add("A103");
12
13        System.out.println("Queue : " + q);
14    }
15 }
```

Input

stdin input

Output

```
Queue : [A101, A102, A103]
Served : A101
Next Customer : A102
Remaining Queue : [A102, A103]
```

2

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name and ID. The assignment is titled "CO2-AT2 Assignment". On the left, a "QUESTIONS" sidebar lists three questions, with the second one, "Q2 Railway Coach and Seat Parser", selected. The main area displays the question details: "Indian Railways stores ticket information as S5-43. Write a Java program that: Splits the value using split(), Extracts the coach number using substring(), Displays the coach type using charAt()". Below this, the "Your Code" section shows a Java program that splits a ticket string "S5-43" and extracts the coach number and type. The "Input" section has a text box for "stdin input". The "Output" section shows the program's output: "Ticket : S5-43, Seat Number : 43, Coach Number : 5, Coach Type : S".

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CO2-AT2 Assignment

QUESTIONS

- Q1 Bank Token Management System 10 marks
- Q2 Railway Coach and Seat Parser 10 marks
- Q3 Employee ID Normalization System 10 marks

3/3 answered

Q2 Railway Coach and Seat Parser 10 marks

Indian Railways stores ticket information as S5-43. Write a Java program that:

- Splits the value using split()
- Extracts the coach number using substring()
- Displays the coach type using charAt()

Your Code

```
1 class Railway {
2
3     public static void main(String args[]) {
4
5         String ticket = "S5-43";
6
7         String s[] = ticket.split("-");
8
9         String coach = s[0].substring(1);
10
11        char type = ticket.charAt(0);
12
13        System.out.println("Ticket : " + ticket);
14        System.out.println("Seat Number : " + s[1]);
15    }
16 }
```

Input

stdin input

Output

```
Ticket : S5-43
Seat Number : 43
Coach Number : 5
Coach Type : S
```

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 SELVA PRABHU A K
192421234RA

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QUESTIONS

Q1 Bank Token Management System 10 marks ✓

Q2 Railway Coach and Seat Parser 10 marks ✓

Q3 Employee ID Normalization System 10 marks ✓

3/3 answered

Q3 Employee ID Normalization System 10 marks

An HR department receives employee IDs in mixed formats such as emp1023. Create a Java program to:
Remove spaces using trim()
Convert the alphabetic prefix to uppercase using toUpperCase()
Join the prefix and number using concat()

Your Code

```
1 class Employee {  
2  
3     public static void main(String args[]) {  
4  
5         String id = " emp1023 ";  
6  
7         id = id.trim();  
8  
9         String prefix = id.substring(0, 3).toUpperCase();  
10  
11        String number = id.substring(3);  
12  
13        String newId = prefix.concat(number);  
14    }
```

Input
stdin input

Output
Original ID : emp1023
New ID : EMP1023

